When Does Market Access Become Local Development?

Platform Integration, Comparative-Advantage Realization, and Local Value Capture

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Abstract

Platform integration can lower consumer prices everywhere without generating local production and income everywhere. This paper develops a tractable theory of that divergence. A platform simultaneously reallocates household spending from a locally embedded channel toward an external channel and expands the national market accessible to local producers. New Structural Economics disciplines the supply response: regional endowments determine latent comparative advantage; infrastructure determines whether that advantage survives actual transaction costs and supports viable entry; and a fixed-cost–variable-cost choice determines whether production relies on an external platform organization or becomes embedded in local complementary services. Payment flows, rather than an assumed local-capability index, generate both consumer-channel retention and producer-side value capture. The model yields four results. Stronger latent comparative advantage lowers the infrastructure thresholds for entry and local organizational embeddedness. Nominal and consumption-equivalent local income have distinct infrastructure thresholds; the latter is weakly lower because consumers also benefit from the price decline. Government participation that relaxes a shared-infrastructure financing constraint raises infrastructure supply and reduces the participation needed to cross those thresholds. Facilitation is not equivalent to protection: protection can induce localization only by worsening the external channel, thereby reducing consumer access and possibly deterring marginal producers. Illustrative numerical exercises establish that the relevant parameter regions are nonempty; they are not a calibration.

Keywords: platform integration; local value capture; latent comparative advantage; actual comparative advantage; facilitating state; regional development.
JEL codes: F12, O14, O18, R11, L81.

1 Introduction

Digital platforms and domestic market integration can make a national market look more uniform from the consumer's side. A household in a remote place can buy varieties previously available only in a large city, and the effective price of those varieties falls as search, payment, and delivery improve. Evidence from online trade in China shows that e-commerce reduces the role of distance and disproportionately improves consumption access in smaller and more remote cities (Fan et al., 2018). Rural e-commerce expansion also reduced the cost of living for adopters in the experiment studied by Couture et al. (2021). Similar consumer gains arise when modern retail channels enter developing markets (Atkin et al., 2018).

Consumer access, however, is not the same object as local development. The rural e-commerce program studied by Couture et al. (2021) produced little average evidence of higher income for local producers and workers. Modern retail entry can lower the cost of living while displacing income and profits in incumbent local stores (Atkin et al., 2018). More generally, reductions in domestic trade costs do not guarantee that peripheral places industrialize: highway connections to larger markets reduced industrial growth in the peripheral Chinese counties examined by Faber (2014). These findings are not evidence that integration is generally harmful. They show that an improvement in the set of goods a place can buy need not coincide with an improvement in what that place can competitively produce, organize, and retain.

This paper asks when platform-enabled market access becomes local development. Its central distinction is between *access* and *capture*. Access describes the effective prices and market opportunities faced by local consumers and producers. Capture describes the location of the production, service, and profit income generated by the associated transactions. A platform can improve inbound consumer access while shifting local expenditure away from locally embedded wholesale, retail, logistics, and after-sales services. The same platform can improve outbound producer access, but a local industry must still be sufficiently competitive to enter, and it may continue to purchase marketing, fulfillment, certification, settlement, and supply chain services from outside the region after entry. Market participation therefore admits an economically important intermediate state: local production has been realized, but its organization remains externally dependent.

New Structural Economics (NSE) supplies the production-side discipline needed to explain this intermediate state. NSE starts from the proposition that an economy's viable industrial structure is conditioned by its endowment structure and that each industrial structure requires corresponding hard and soft infrastructure (Lin, 2011, 2012). A firm following the comparative advantage implied by local factor endowments can nevertheless fail to operate if actual transaction costs are too high. Lin and Wang (2023) formalize this distinction as one between latent comparative advantage (LCA), based on relative production costs, and actual comparative advantage (ACA), based on relative total costs after transaction costs. The distinction matters for policy because infrastructure constraints often cannot be removed by a single firm. A facilitating state can help industries with potential comparative advantage overcome shared infrastructure and coordination constraints (Lin and Monga, 2010); it is not meant to sustain indefinitely an industry that remains nonviable in open competition (Lin, 2003).

The use of NSE here is deliberately narrower than a general theory of structural transformation and broader than relabeling an exogenous "capacity" parameter. Endowments determine relative production costs. Infrastructure affects actual delivered costs and producer market access. Those objects jointly determine viable entry and the organizational form used after entry. Organizational choice then determines the location of complementary-service payments. Thus NSE affects both whether local production exists and how much of the resulting value is locally embedded. It does not replace consumer demand, the platform shock, or the local-income propagation mechanism.

The model has five parts. First, a representative household consumes a market good through a local channel and an external platform channel. Platform integration lowers the platform price, raises its expenditure share, and lowers the market-good price index. The local income content of each channel comes from a payment-flow table. When the local channel has greater local income content, the channel shift creates a consumer-side capture drag.

Second, a focal region is embedded in an otherwise exogenous national market. A competitive background industry determines local factor opportunity costs. The region's capital–labor endowment and the candidate industry's factor intensity generate its unit production cost. LCA is defined strictly as a region–industry double relative production cost. ACA, viability, and realization are separately defined: ACA includes actual transaction costs, viability requires nonnegative profit without persistent protection, and realization is equilibrium entry and sales.

Third, a representative candidate producer faces CES national demand. The same platform-integration index that reduces the consumer platform price also expands producer market access. Shared infrastructure reduces outbound iceberg costs, and platform integration makes that infrastructure more productive. A producer can rely on an external platform organization with a low fixed cost and a high unit service payment, or build a locally embedded organization with a higher fixed cost and a lower unit organizational cost. This conditional fixed-cost–variable-cost ordering is related to the trade intermediation mechanism in Ahn et al. (2011), but the present paper uses a representative industry rather than a heterogeneous-firm distribution.

Fourth, the producer's revenue is split according to the location of actual payments. In the external mode, the unit platform-service payment leaves the region. In the local mode, production and complementary organizational services are locally supplied. These payment flows generate the producer-retention rate; no separate retention-capacity function is assumed. The resulting first-round producer income enters an EL-compatible local-income propagation equation. The propagation block is a one-region, household-endogenous multiplier, closely related to standard social-accounting and local-multiplier reasoning (Miller and Blair, 2009; Moretti, 2010). Its role is conditional: it transmits first-round income generated by the new model rather than constituting the new contribution.

Finally, a reduced-form infrastructure provider faces a financing wedge that falls with government participation. This block follows the enabling-state logic of Lin and Wang (2023): the state helps relax a financing constraint on transaction-cost-reducing infrastructure. It is not a national social planner, and the paper does not claim that the chosen government participation is nationally optimal.

Four propositions organize the results. Proposition 1 shows that stronger LCA lowers the infrastructure required for viable entry. Proposition 2 shows that a second threshold determines organizational embeddedness; stronger LCA also lowers this threshold, and crossing it changes local service income, external payments, and the producer-retention rate. Proposition 3 combines these supply responses with the consumer channel shift. It establishes unique minimum infrastructure thresholds for nonnegative nominal and consumption-equivalent income effects under endpoint-crossing conditions. The real-income threshold is weakly below the nominal-income threshold and is strictly below it when the two roots are interior to the same continuous organizational regime. This qualification matters because discrete entry or organization switches can make the thresholds coincide. Proposition 4 maps the thresholds into minimum government-participation levels and distinguishes facilitation from protection.

The paper contributes to three literatures. Relative to the literature on e-commerce and retail access, it explains why consumer price gains can be more geographically diffuse than producer and income gains. Relative to NSE, it introduces platform-mediated organizational dependence as a distinct stage between realizing production and embedding complementary services locally. Relative to local-multiplier models, it derives the first-round local-income term and retention rates from comparative advantage, entry, organization, and payment location. The contribution is not the multiplier itself and not a claim that local retention should be maximized at the expense of consumers.

The analysis is intentionally a small-region partial equilibrium. It does not contain migration, land, endogenous national expenditure, platform strategic pricing, or New Economic Geography agglomeration. It therefore cannot evaluate national welfare or long-run spatial equilibrium. This restriction keeps the paper focused on the question that motivates it: when a common platform improves access, what structural conditions allow a region to convert that opportunity into locally captured income?

2 The Access–Capture Benchmark

2.1 Households and consumer access

A representative household allocates nominal income among a market-good composite Q_M, a locally produced nontraded good N, and an outside numeraire Z:

$$U = Q_M^{\alpha_M} N^{\beta} Z^{1-\alpha_M-\beta}, \qquad \alpha_M > 0, \quad \beta \geq 0, \quad \alpha_M + \beta < 1. \tag{1}$$

Following the standard CES benchmark (Dixit and Stiglitz, 1977), the market-good composite is purchased through a locally embedded channel A and an external platform channel P:

$$Q_M = \left[(1-a_P)^{1/\eta} Q_A^{(\eta-1)/\eta} + a_P^{1/\eta} Q_P^{(\eta-1)/\eta}\right]^{\eta/(\eta-1)}, \qquad \eta > 1. \tag{2}$$

Normalize $p_A = 1$. Platform integration z lowers the effective platform price:

$$p_P(z) = \bar{p}_P e^{-z}. \tag{3}$$

This price can include the delivered product price, search and payment costs, and any consumer-facing platform charge. It is distinct from the producer-side organizational payment introduced below.

The platform expenditure share and market-good price index are:

$$s(z) = \frac{a_P p_P(z)^{1-\eta}}{(1-a_P)p_A^{1-\eta} + a_P p_P(z)^{1-\eta}}, \tag{4}$$

$$P_M(z) = \left[(1-a_P)p_A^{1-\eta} + a_P p_P(z)^{1-\eta}\right]^{1/(1-\eta)}. \tag{5}$$

Standard CES differentiation gives:

$$s'(z) = (\eta-1)s(z)[1-s(z)] > 0, \qquad \frac{d\ln P_M(z)}{dz} = -s(z) < 0. \tag{6}$$

Thus the platform unambiguously improves the model's consumer-access object: its expenditure share rises and the cost of a unit of Q_M falls.

2.2 Consumer-channel payment incidence

Access does not determine where income is generated. Let ℓ_A and ℓ_P denote the local income generated per unit of expenditure through the two consumer channels. Table 1 clarifies their content. Both measures use the same transaction boundary and therefore can be combined without treating an external payment as a national resource loss.

The baseline assumes $\ell_A > \ell_P$. This is a transparent spatial incidence assumption, not an assertion that platforms always retain less value locally. A platform that develops local fulfillment, seller services, or ownership can have $\ell_P \geq \ell_A$; that reverse case is reported in the appendix.

The local income content of household market-good expenditure is:

$$\omega_C(z) = [1-s(z)]\ell_A + s(z)\ell_P. \tag{7}$$

Define:

$$D(z) = 1 - \beta - \alpha_M \omega_C(z). \tag{8}$$

The maintained stability condition is $D(z) > 0$. When $\ell_A > \ell_P$, platform substitution lowers ω_C

Table 1: Consumer-channel payment incidence

Payment generated by one unit of local household expenditure	Local channel A	External platform P
Local wholesale, retail, storage, and after-sales value added	local income	limited in baseline
External production and interregional shipping	external payment	external payment
Platform, settlement, and externally supplied digital services	limited	external payment
Locally owned operating surplus	local income	included if locally owned
Local income content	ℓ_A	ℓ_P

and raises D. The corresponding proportional capture drag is:

$$\Lambda(z) = \frac{\alpha_M(\ell_A - \ell_P)s'(z)}{D(z)} > 0. \tag{9}$$

The numerator records how rapidly expenditure moves toward the lower-retention channel; the denominator captures repeated local spending. This term is consumer-side channel displacement. It is separate from the producer-side external platform payment developed in Section 4.

3 Endowments, Comparative Advantage, and Producer Access

3.1 A focal region and factor opportunity costs

Consider a small focal region i with baseline factor endowments K_i and L_i. Let:

$$e_i = \frac{K_i}{L_i}. \tag{10}$$

A competitive background industry 0 uses:

$$Y_{i0} = A_0 K_{i0}^{\theta_0} L_{i0}^{1-\theta_0}. \tag{11}$$

The candidate industry is small relative to the background economy. The background industry therefore determines the factor opportunity costs relevant for its entry decision. Over the range studied, marginal factor services can be supplied at those opportunity costs without changing the factor-price schedule. This assumption can represent underused capacity, additional work hours, or a locally elastic short-run supply margin. With the background good as numeraire, competitive factor pricing gives:

$$w_i = (1-\theta_0)A_0 e_i^{\theta_0}, \qquad r_i = \theta_0 A_0 e_i^{\theta_0 - 1}. \tag{12}$$

This closure avoids treating factor prices as free parameters while preserving a tractable small-region problem. It is not a full-employment two-sector general equilibrium: a long-run welfare analysis would have to subtract production forgone in the background industry and solve the factor reallocation explicitly.

Candidate industry j uses:

$$Y_{ij} = A_j K_{ij}^{\theta_j} L_{ij}^{1-\theta_j}. \tag{13}$$

Its unit production cost is:

$$\begin{aligned} c_{ij}^P &= \frac{1}{A_j} \left(\frac{r_i}{\theta_j}\right)^{\theta_j} \left(\frac{w_i}{1-\theta_j}\right)^{1-\theta_j} \\ &= \xi_j e_i^{\theta_0-\theta_j}, \qquad \xi_j > 0. \end{aligned} \tag{14}$$

If the candidate industry is more capital intensive than the background industry, $\theta_j > \theta_0$, a higher regional capital–labor endowment lowers its relative unit cost. This is the minimal factor-proportions channel required for an NSE paper. It is consistent with the broader result that factor abundance shifts production and trade toward industries intensive in the abundant factor (Romalis, 2004), and with the NSE view that industrial composition evolves with endowment structure (Ju et al., 2015).

3.2 LCA, ACA, viability, and realization

The four concepts used below are related but not interchangeable.

Definition 1 (Latent comparative advantage)**.** Let h be an external reference region and 0 a background industry in each region. Region i's production-cost comparative-advantage index for industry j is:

$$\mathcal{C}_{ij}^P = \frac{c_{ij}^P / c_{i0}^P}{c_{hj}^P / c_{h0}^P}. \tag{15}$$

Region i has latent comparative advantage in j when $\mathcal{C}_{ij}^P < 1$.

The double ratio is essential. A low absolute unit cost is not by itself a definition of comparative advantage. In the analytical and numerical comparative statics below, c is a monotone sufficient statistic for $\mathcal{C}_{ij}^P$, holding the reference-region and background-industry ratios fixed:

$$\frac{\partial c}{\partial \mathcal{C}_{ij}^P} > 0. \tag{16}$$

Lower c therefore means stronger LCA, but the underlying definition remains equation (15).

Definition 2 (Actual comparative advantage)**.** Actual comparative advantage compares the same region–industry double relative cost after adding the transaction costs generated by the current hard and soft infrastructure. It is present when the minimum delivered-cost ratio for industry j, relative to industry 0, is below the corresponding ratio in region h.

To make this definition operational, let the outbound iceberg cost be:

$$\tau_O(z,G) = \bar{\tau}_O \exp[-(\gamma + \chi z)G], \qquad \bar{\tau}_O > 1, \quad \gamma \geq 0, \quad \chi > 0, \tag{17}$$

and restrict the local analysis to the range in which $\tau_O(z,G) \geq 1$. The baseline component $\bar{\tau}_O^{1-\sigma}$ will be absorbed into the market-access normalization below. The delivered marginal cost under organization r is $\tau_O(z,G)m_r$, where m_r is defined below. Holding the external reference fixed, a larger G lowers the local total-cost ratio. Thus a region can possess LCA while lacking ACA because inadequate infrastructure leaves its total delivered cost too high.

Definition 3 (Viability and realization)**.** The candidate producer is viable if at least one available organization earns nonnegative profit in open competition without persistent protection:

$$\max\{\pi_P, \pi_L\} \geq 0.$$

Comparative advantage is realized when the producer actually enters, produces, and sells in equilibrium.

LCA describes a structural cost potential; ACA describes that potential under actual transaction conditions; viability is a firm-level profit condition; realization is an equilibrium outcome. Observed exports or revealed comparative advantage can measure realization, but they do not define LCA.

3.3 Producer market access

The candidate producer faces a national CES demand schedule:

$$q_r = \mathcal{M}(z,G)p_r^{-\sigma}, \qquad \sigma > 1. \tag{18}$$

Let $\mathcal{M}_0 e^{\varepsilon z}$ be the national expenditure opportunity available to the focal industry after absorbing the invariant factor $\bar{\tau}_O^{1-\sigma}$ into $\mathcal{M}_0$. Substituting the variable component of equation (17) into CES demand gives:

$$\mathcal{M}(z,G) = \mathcal{M}_0 \exp\left[\varepsilon z + (\sigma - 1)(\gamma + \chi z)G\right]. \tag{19}$$

This formulation has the standard market-access structure in which bilateral trade costs enter with elasticity $1 - \sigma$ (Donaldson and Hornbeck, 2016). The term εz captures the direct producer-access effect of platform integration. The cross term $(\sigma - 1)\chi z G$ captures complementarity: logistics, certification, quality control, digital connectivity, and similar infrastructure have a larger payoff when a platform can connect the producer to a broader market.

Equation (19) is not the consumer price equation written twice. Consumer access in equation (3) concerns what local households can buy; producer access concerns the demand that local producers can reach. The same reform index z moves both, but their incidence and dependence on G differ.

3.4 Entry and the realization threshold

The producer can use an external platform organization P or a locally embedded organization L:

$$m_P = c + d_P, \qquad m_L = c + d_L, \qquad 0 \leq d_L < d_P, \tag{20}$$

$$F_P = f, \qquad F_L = f + F, \qquad f > 0, \quad F > 0. \tag{21}$$

The external organization uses a platform's existing national marketing, settlement, fulfillment, and supply-chain system. It therefore avoids the shared fixed cost F, but it entails a larger unit service payment. The local organization builds or coordinates those complementary services locally. This ordering is conditional, not a universal property of digital platforms. Its purpose is to represent a well-defined organizational choice: buy an external service flow or incur a fixed cost to embed the service locally.

With constant elasticity demand, the markup, price, revenue, and profit are:

$$\mu = \frac{\sigma}{\sigma - 1}, \qquad p_r = \mu m_r, \tag{22}$$

$$\mathcal{R}_r(z, G) = \mathcal{M}(z, G)\mu^{1-\sigma} m_r^{1-\sigma}, \qquad \pi_r(z, G) = \frac{\mathcal{R}_r(z, G)}{\sigma} - F_r. \tag{23}$$

The producer selects:

$$r^*(z, G) \in \arg\max\{0, \pi_P(z, G), \pi_L(z, G)\}. \tag{24}$$

Define:

$$k = \frac{\mu^{1-\sigma}}{\sigma}, \qquad \Omega(z) = (\sigma - 1)(\gamma + \chi z) > 0. \tag{25}$$

The zero-profit infrastructure threshold for organization r is:

$$G_r^0(z, c) = \frac{\ln\left[\frac{F_r}{k\mathcal{M}_0 e^{\varepsilon z}(c + d_r)^{1-\sigma}}\right]}{\Omega(z)}. \tag{26}$$

The economically relevant entry threshold is:

$$G_E(z, c) = \max\left\{0, \min[G_P^0(z, c), G_L^0(z, c)]\right\}. \tag{27}$$

Proposition 1 (Comparative-advantage realization)**.** *Suppose $\Omega(z) > 0$. A unique minimum infrastructure level G_E supports viable entry. At an interior threshold:*

$$\frac{\partial G_E}{\partial c} > 0, \qquad \frac{\partial G_E}{\partial \mathcal{C}_{ij}^P} > 0. \tag{28}$$

Hence stronger LCA lowers the infrastructure required to obtain ACA and viable realization. If the raw threshold is negative, the economically truncated threshold is zero and the comparative static is weak.

The result is not an assumption that LCA raises local income. It follows from the profit problem: a lower endowment-conditioned production cost raises operating profit under either organization and reduces the amount of transaction-cost-reducing infrastructure needed to cover the fixed cost. Proofs of all propositions are in the appendix.

4 Organization, Capture, and Local Income

4.1 The organizational-embeddedness threshold

The local mode has a higher operating-profit coefficient because $d_L < d_P$, but it must cover the additional fixed cost F. The infrastructure level at which $\pi_L = \pi_P$ is:

$$G_X(z,c) = \frac{\ln\left[\dfrac{F}{k\mathcal{M}_0 e^{\varepsilon z}\{(c+d_L)^{1-\sigma} - (c+d_P)^{1-\sigma}\}}\right]}{\Omega(z)}. \tag{29}$$

The economically relevant embeddedness threshold is:

$$G_L(z,c) = \max\{0, G_X(z,c)\}. \tag{30}$$

The full state classification requires distinguishing G_P^0, G_L^0, and G_X. Let:

$$b_r = k(c+d_r)^{1-\sigma}, \qquad \bar{F} = f\frac{b_L - b_P}{b_P}. \tag{31}$$

If $F > \bar{F}$, then:

$$G_P^0 < G_L^0 < G_X. \tag{32}$$

The equilibrium consequently passes through three regions as infrastructure rises:

$$\text{no entry} \quad \longrightarrow \quad \text{external platform dependence} \quad \longrightarrow \quad \text{local organizational embeddedness.}$$

If $F \leq \bar{F}$, the platform-dependent middle region can disappear: the producer may move directly from no entry to the local organization. The middle state is therefore an economic result with a transparent parameter condition, not a stage imposed by terminology.

4.2 Producer-side payment incidence

Producer revenue is not automatically local income. Let $d_P q_P$ be the payment to the external platform and externally supplied complementary services. Production factors and the residual operating surplus are locally owned in the baseline. Under the local mode, the organizational-service payment $d_L q_L$ is made to local providers. Table 2 records the incidence.

Fixed costs affect viability and organization, but locally supplied fixed-cost services are payments to local factors or firms. Deducting them again from local income would double count a local

Table 2: Producer organization and spatial payment incidence

Payment generated by candidate-industry sales	External mode P	Local mode L
Local production-factor income	local	local
Local producer operating surplus	local	local
Unit marketing, settlement, fulfillment, and supply-chain service	external $d_P q_P$	local $d_L q_L$
Shared organizational fixed cost	not incurred	locally supplied F
Common entry cost	locally supplied f	locally supplied f
External payment included in retention measure	$d_P q_P$	0

resource payment. Conversely, the external unit payment is excluded from local income even if it is not a national loss.

Let B_j^r denote first-round local income generated by candidate-industry sales:

$$B_j^r(z, G) = \rho_r \mathcal{R}_r(z, G). \tag{33}$$

The payment table implies:

$$\rho_P = 1 - \frac{d_P}{\mu(c + d_P)} < 1, \qquad \rho_L = 1. \tag{34}$$

If a fraction $\lambda \in [0, 1]$ of platform services is supplied locally:

$$\rho_P(\lambda) = 1 - \frac{(1-\lambda)d_P}{\mu(c + d_P)}. \tag{35}$$

The baseline is $\lambda = 0$. This robustness parameter changes payment location, not productive efficiency.

Proposition 2 (Organizational embeddedness)**.** *Suppose $d_L < d_P$, $F > 0$, and $\Omega(z) > 0$. There exists a minimum embeddedness threshold G_L. At an interior threshold:*

$$\frac{\partial G_L}{\partial c} > 0, \qquad \frac{\partial G_L}{\partial \mathcal{C}_{ij}^P} > 0. \tag{36}$$

For $G \geq G_L$, the producer selects L, and the switch satisfies:

$$\rho_L > \rho_P, \qquad \mathcal{R}_L > \mathcal{R}_P, \qquad B_j^L > B_j^P. \tag{37}$$

If $F > \bar{F}$, platform dependence is a distinct equilibrium region between entry and local embeddedness. If $F \leq \bar{F}$, that region may disappear.

This proposition is the main way NSE affects capture beyond production. Stronger LCA makes local sales sufficiently large to cover the shared organizational fixed cost at a lower infrastructure level. Crossing the threshold does not merely relabel the firm: it reduces external payments and creates local complementary-service income.

4.3 Nominal and consumption-equivalent local income

Let $B_0 > 0$ denote autonomous first-round income from the background economy over the local partial-equilibrium range. Consistent with the locally elastic factor-supply assumption above, candidate-industry factor payments are incremental nominal receipts; their full opportunity cost is not a welfare gain. Household spending generates additional rounds of local income. The share β spent on locally produced nontraded goods remains local. The share α_M spent on market goods generates local income at rate $\omega_C(z)$. Avoiding any duplication of the first-round producer payments, nominal local income satisfies:

$$Y = B_0 + B_j^{r^*}(z, G) + \beta Y + \alpha_M \omega_C(z) Y. \tag{38}$$

Therefore:

$$Y(z, G) = \frac{B_0 + B_j^{r^*}(z, G)}{D(z)}. \tag{39}$$

This is a conditional local-income accounting equilibrium. It is appropriate when the relevant local services and nontraded goods can respond at a fixed short-run opportunity cost. It is not a full welfare measure.

Consumption-equivalent income deflates nominal income by the market-good price index:

$$R(z, G) = \frac{Y(z, G)}{P_M(z)^{\alpha_M}}. \tag{40}$$

Define the candidate industry's share in first-round local income:

$$\theta_r(z, G) = \frac{B_j^r(z, G)}{B_0 + B_j^r(z, G)}. \tag{41}$$

Within a fixed organizational state, equations (19), (9), and (39) yield:

$$\frac{d \ln Y}{dz} = \theta_r(z, G) \left[\varepsilon + (\sigma - 1)\chi G\right] - \Lambda(z). \tag{42}$$

The first term is the local producer-income gain from better platform access. It is larger when the candidate industry already has a greater weight in local income and when infrastructure complements platform access. The second term is the consumer-channel capture drag.

The consumption-equivalent effect is:

$$\frac{d \ln R}{dz} = \frac{d \ln Y}{dz} + \alpha_M s(z). \tag{43}$$

The final term is the direct consumer price benefit. Equations (42) and (43) preserve the original access–capture question while replacing an exogenous producer gain and retention capacity with equilibrium entry, organization, and payment flows.

Proposition 3 (Nominal and real-income thresholds)**.** *Maintain $D(z) > 0$, $\chi > 0$, and the baseline payment ordering. Within each active producer organization, both $d \ln Y/dz$ and $d \ln R/dz$ are strictly increasing in G; in the no-entry state they are constant in G. Entry and the switch from P to L generate upward jumps. If the relevant effect is negative at the lower endpoint and positive at an upper endpoint, there is a unique minimum threshold:*

$$G_Y = \inf \left\{ G : \frac{d \ln Y}{dz} \geq 0 \right\}, \tag{44}$$

$$G_R = \inf \left\{ G : \frac{d \ln R}{dz} \geq 0 \right\}. \tag{45}$$

Pointwise,

$$G_R \leq G_Y. \tag{46}$$

If both roots are interior to the same continuous organizational region,

$$G_R < G_Y. \tag{47}$$

If both effects cross zero at the same discrete entry or organization switch, the thresholds can coincide. Stronger LCA weakly lowers both thresholds and strictly lowers them when the relevant roots remain interior to the same organizational states.

The proposition separates three regional outcomes. Below G_R, producer gains are too small to offset both channel displacement and the relevant cost of living, so nominal and consumption-equivalent income fall. Between G_R and G_Y, nominal income falls but the consumer price gain is large enough to raise consumption-equivalent income. Above G_Y, producer access and local capture are strong enough for both measures to rise. The classification is conditional: if $\ell_P \geq \ell_A$, capture drag is nonpositive and the low-infrastructure loss region can disappear.

5 The Facilitating State

5.1 Infrastructure supply

The preceding analysis treats G as a local structural condition. This section endogenizes its supply without introducing a national social planner. Let $\vartheta \in [0, 1]$ denote government participation in infrastructure finance. The financing wedge is:

$$\kappa_I(\vartheta) = \kappa_p - (\kappa_p - \kappa_g)\vartheta, \qquad 0 < \kappa_g < \kappa_p. \tag{48}$$

The infrastructure provider solves:

$$\max_{G \geq 0} \left\{ \mathcal{U}_I(G) - \kappa_I(\vartheta) C(G) \right\}, \tag{49}$$

where:

$$\mathcal{U}_I' > 0, \quad \mathcal{U}_I'' < 0, \quad C' > 0, \quad C'' \geq 0. \tag{50}$$

$\mathcal{U}_I$ is the provider's appropriable revenue from infrastructure services. The gap between this revenue and the broader producer-access gains captures the shared nature of logistics, standards, certification, digital systems, and other infrastructure. Government participation lowers the financing wedge; it does not directly subsidize candidate-industry output.

An interior solution satisfies:

$$\mathcal{U}_I'[G(\vartheta)] = \kappa_I(\vartheta) C'[G(\vartheta)]. \tag{51}$$

The curvature assumptions guarantee uniqueness.

5.2 Government-participation thresholds

Proposition 4 (State enabling and protection)**.** *The infrastructure supply in equation* (49) *is unique and:*

$$G'(\vartheta) > 0. \tag{52}$$

For $H \in \{E, L, R, Y\}$, define the minimum government participation needed to cross an existing threshold:

$$\vartheta_H = \inf\{\vartheta : G(\vartheta) \geq G_H\}. \tag{53}$$

Greater government participation reduces each gap $G_H - G(\vartheta)$. Stronger LCA weakly lowers $\vartheta_E, \vartheta_L, \vartheta_R$, and ϑ_Y, with strict effects where the corresponding structural thresholds are interior.

Facilitation and protection are not equivalent. Facilitation raises G, reduces actual transaction constraints, and expands producer market access without raising the consumer platform price. A protection instrument that raises the external consumer price and the external producer organization cost can make L relatively more attractive, but it worsens consumer access and raises the platform-entry threshold for a marginal producer.

To see the second part formally, let protection multiply p_P by a wedge $\tau_C > 1$ and raise d_P. The consumer price index rises and the platform expenditure share falls. On the producer side:

$$\frac{\partial G_P^0}{\partial d_P} = \frac{\sigma - 1}{\Omega(z)(c + d_P)} > 0. \tag{54}$$

Protection therefore makes external-platform entry harder. At the same time, raising d_P increases the local mode's variable-cost advantage and can lower G_X. It can induce localization by degrading the alternative. Facilitation instead allows the region to cross the entry and embeddedness thresholds by reducing a real shared constraint. This is the model's precise version of the NSE distinction between facilitating a potentially viable industry and protecting a nonviable activity.

The state result is deliberately limited. It does not prove that more government participation is always desirable, that the provider captures all social returns, or that $\vartheta = 1$ is optimal. A complete national welfare problem would require government fiscal costs, external platform profits, other regions, and general-equilibrium factor responses, all of which are outside the focal-region model.

6 Illustrative Numerical Exercises

6.1 Purpose and parameterization

The numerical exercises have one purpose: to demonstrate that the parameter regions described by Propositions 1–4 are nonempty and to display the model's discontinuities. They are not a calibration and none of the values is presented as an estimate for China. Table 3 reports the standardized baseline.

Table 3: Illustrative baseline parameters

Block	Parameters	Values
Household	α_M, β, η	0.35, 0.30, 6
Consumer channels	ℓ_A, ℓ_P, B_0	0.70, 0.10, 1
Producer demand	$\sigma, \mathcal{M}_0, \varepsilon$	4, 5, 0.05
Infrastructure access	γ, χ	0.10, 0.20
Production and organization	c, d_P, d_L, f, F	0.8, 0.4, 0.1, 0.3, 0.5
State block	κ_p, κ_g, a_I	2.0, 0.5, 0.8

The consumer CES weight is $a_P = 0.4$, and the pre-integration platform price is $\bar{p}_P = 1.2$. For the state exercise:

$$\mathcal{U}_I(G) = a_I \ln(1 + G), \qquad C(G) = \frac{G^2}{2}. \tag{55}$$

Equation (51) then has the closed-form solution:

$$G(\vartheta) = \frac{-1 + \sqrt{1 + 4a_I/\kappa_I(\vartheta)}}{2}. \tag{56}$$

6.2 The structural phase diagram

Figure 1 varies the candidate industry's relative production cost and infrastructure at $z = 0.5$. Moving right weakens LCA. Moving up improves the infrastructure through which LCA can be converted into viable entry and local organization.

The gray region contains no viable candidate industry. The blue region contains local production that reaches the national market through the external platform organization. The orange region contains locally embedded production and complementary services. At a given G, a stronger LCA industry is more likely both to enter and to embed. At a given production cost, infrastructure can

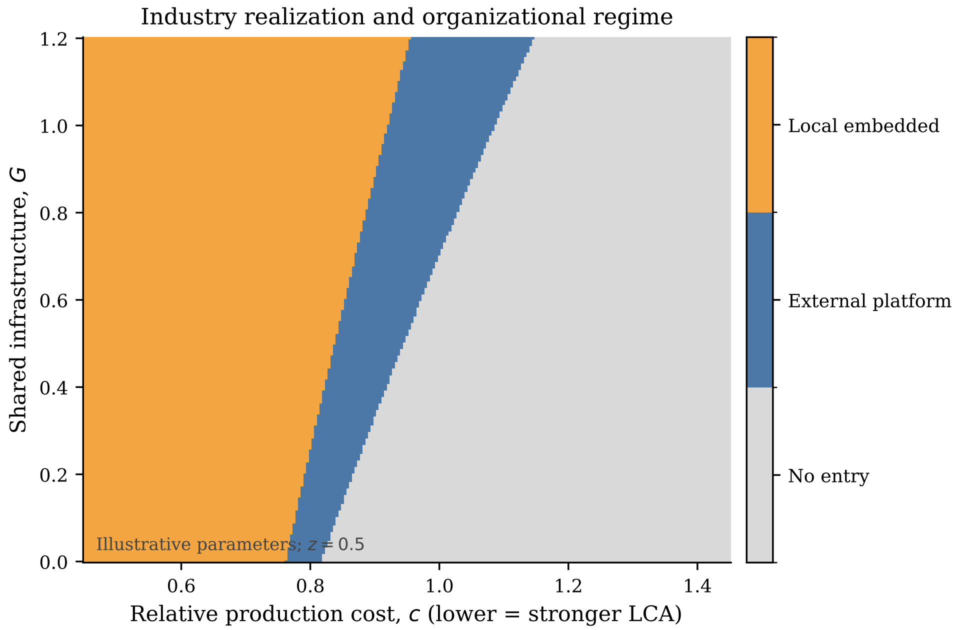

Figure 1: Industry realization and organizational regime

move the region through all three states. The middle region is sizable under the baseline because $F > \bar{F}$.

6.3 Development thresholds

Figure 2 plots the entry, organizational, real-income, and nominal-income thresholds against c. The curves are computed from the exact equilibrium choice and the minimum-G definitions in equations (27), (30), (44), and (45).

All four thresholds weakly rise as LCA weakens. Kinks are economically meaningful: they occur when the identity of the active entry mode or the organizational state containing an income threshold changes. In the baseline at $c = 0.8, z = 0.5$, the raw state thresholds are:

$$G_P^0 = -0.0702, \qquad G_L^0 = 0.1261, \qquad G_X = 0.2561.$$

The real- and nominal-income thresholds are:

$$G_R = 0.0463, \qquad G_Y = 0.5505.$$

The numerical ordering is strict, although the formal proposition preserves the weaker boundary result when a discrete switch determines both thresholds.

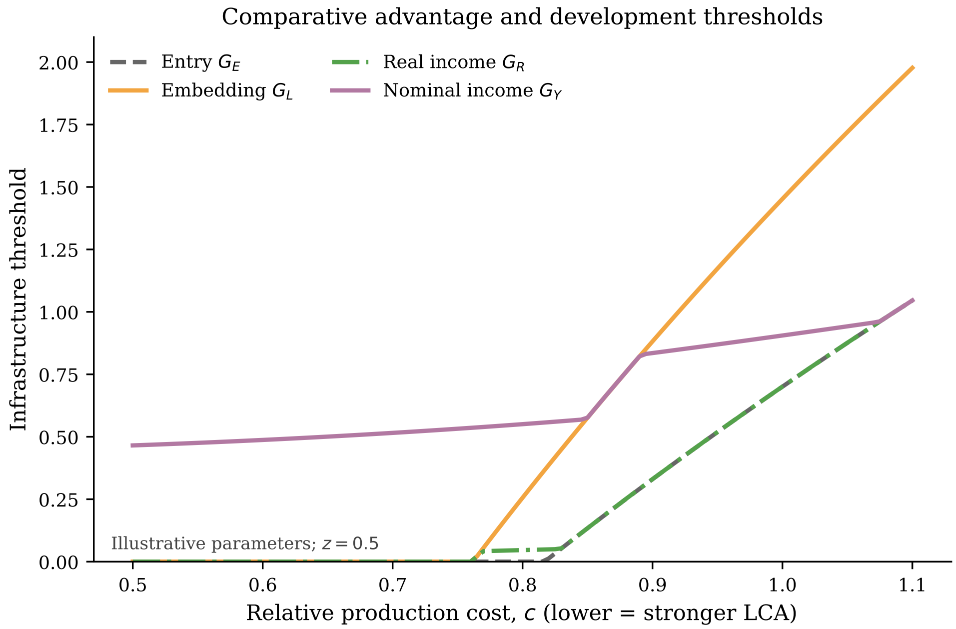

Figure 2: Comparative advantage and development thresholds

6.4 Platform integration under alternative state participation

The state specification yields:

$$G(0.1) = 0.3261, \qquad G(0.5) = 0.4434, \qquad G(0.9) = 0.7169.$$

Figure 3 follows platform integration for those three infrastructure levels. The first panel is common across participation levels because the baseline state intervention affects producer infrastructure, not the consumer platform price. The second and third panels report marginal effects, not income indices.

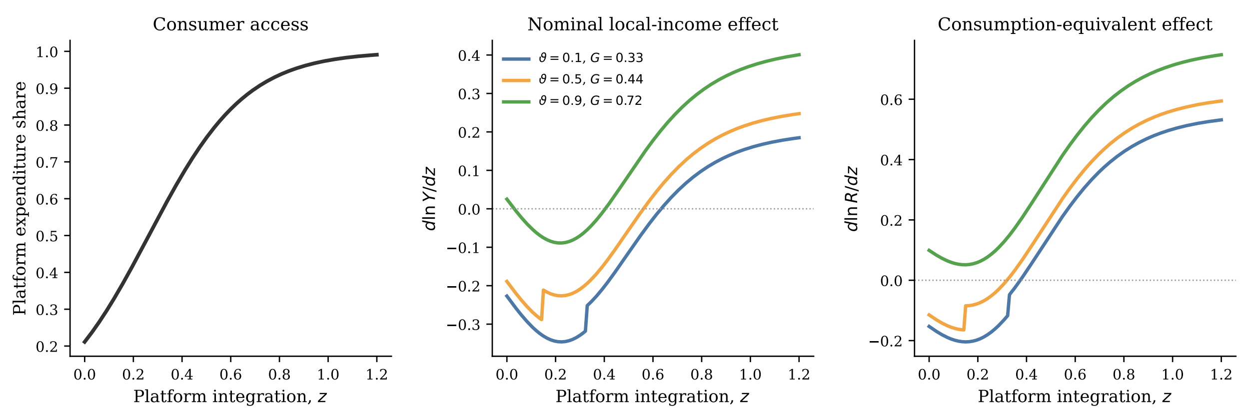

Figure 3: Platform integration, state participation, and local-income effects

Platform penetration rises for every region. At low participation, the nominal effect is initially negative and becomes positive only after producer-access gains and organizational embeddedness strengthen. The consumption-equivalent effect crosses zero earlier because it includes the price gain. Higher participation shifts both effects upward, but the consumer platform share is unchanged. The small jumps in the income-effect curves are the entry or P-to-L organization switches established in the analytical model.

The appendix reports five mechanism closures. Setting $\ell_A = \ell_P$ eliminates consumer-side capture drag. Equating ρ_P and ρ_L eliminates the producer-side spatial payment difference. Setting $\chi = 0$ eliminates infrastructure–platform complementarity. A positive λ partially localizes platform services. Finally, an equal-localization comparison shows that protection and facilitation have different consumer-access and marginal-entry consequences. Every figure is generated from the same public code and is accompanied by CSV source data.

7 Discussion and Testable Implications

7.1 What the model does and does not call development

The model distinguishes three outcomes. P_M measures consumer access. Y measures nominal local income generated within the focal accounting boundary. R measures how much market-good consumption that nominal income can support after the platform price change. None is a national welfare measure. A payment to an external domestic platform is a leakage from the focal region but remains income somewhere in the national economy. Likewise, a high local-retention rate is not itself a sufficient policy objective: a policy can mechanically raise retention by making an efficient external channel more costly.

This distinction prevents a protectionist reading of the access–capture problem. A region with little infrastructure may receive a genuine consumer gain even when nominal local income falls. The relevant development question is not how to preserve every incumbent local intermediary. It is how to enable industries consistent with the region's potential comparative advantage to enter, expand, and build complementary local services without sacrificing the consumer gains from national integration.

7.2 Testable implications

Although the paper contains no original data, its mechanisms generate specific empirical predictions.

First, platform expansion should reduce consumer price dispersion more broadly than it increases local producer income. Consumer effects should depend primarily on pre-platform access and platform adoption, whereas producer-income effects should load on the interaction between pre-existing endowment–industry congruence and infrastructure. This prediction sharpens the access–income divergence documented by Couture et al. (2021).

Second, among locations with similar platform access, lower endowment-conditioned relative production costs should predict earlier seller entry. Infrastructure should have larger producer effects in industries that are intensive in locally abundant factors. A credible test would construct LCA from pre-shock regional endowments and national industry factor intensities, rather than infer comparative advantage from post-shock sales.

Third, producer entry and local capture should not be treated as the same outcome. In the platform-dependent region, local seller sales can rise while platform, settlement, fulfillment, certification, and marketing payments remain external. Crossing the embeddedness threshold predicts a discrete increase in local complementary-service establishments, employment, or payroll relative to sales. Buyer–seller locations alone are therefore insufficient to measure capture; service-provider and ownership locations are also needed.

Fourth, facilitating infrastructure and protection should have different joint signatures. Facilitation should raise outbound seller participation without raising the local consumer platform price. Protection may increase the local share of some services while reducing platform adoption, raising consumer prices, and deterring marginal platform-dependent sellers. Empirically similar local-sales shares can therefore conceal very different access and productivity consequences.

Fifth, partial localization of platform services should compress the difference between P and L. Regions hosting fulfillment, seller operations, payment, or platform-service employment should retain more value for a given volume of platform trade. This prediction separates the location of production from the location of organizational capture.

7.3 Scope and extensions

Several mechanisms are intentionally absent. Firm heterogeneity would turn the two organization thresholds into productivity cutoffs, as in trade intermediation models, but is not needed for the regional thresholds derived here. New Economic Geography feedback could make local service depth and market access endogenous to entry, possibly generating multiple equilibria. Dynamic endowment accumulation could move a region across industries over time. Platform strategic pricing could make the consumer and producer wedges jointly endogenous. These are natural extensions, but adding them to the present paper would obscure its distinct result: platform access can be diffuse while production and organizational capture remain structurally conditional.

8 Conclusion

Platform integration changes both sides of a local economy. It improves what households can buy, and it expands what local producers might sell. Those opportunities do not automatically become local development. Endowments determine latent comparative advantage; infrastructure determines whether that advantage survives actual transaction costs and supports viable entry; and organizational choice determines whether complementary-service payments remain local.

This chain produces a platform-dependent intermediate state in which local production exists but organizational capture remains external. It also produces separate thresholds for nominal and consumption-equivalent local income. Consumer price gains make the real-income threshold weakly lower, while discrete entry or organization switches explain why strict ordering is conditional rather than universal. A facilitating state can help a potentially viable industry cross these thresholds by relaxing shared infrastructure financing constraints. Protection can also induce localization, but only by degrading an external channel and therefore cannot be treated as equivalent.

The central conclusion is consequently not that platforms either help or hurt local development. It is that access and capture are different equilibrium objects. A common platform can equalize consumer access while leaving the production and organizational foundations of local income highly unequal.

References

Ahn, J., Khandelwal, A. K., and Wei, S.-J. (2011). The role of intermediaries in facilitating trade. *Journal of International Economics*, 84(1):73–85.

Atkin, D., Faber, B., and Gonzalez-Navarro, M. (2018). Retail globalization and household welfare: Evidence from mexico. *Journal of Political Economy*, 126(1):1–73.

Couture, V., Faber, B., Gu, Y., and Liu, L. (2021). Connecting the countryside via e-commerce: Evidence from china. *American Economic Review: Insights*, 3(1):35–50.

Dixit, A. K. and Stiglitz, J. E. (1977). Monopolistic competition and optimum product diversity. *American Economic Review*, 67(3):297–308.

Donaldson, D. and Hornbeck, R. (2016). Railroads and american economic growth: A "market access" approach. *Quarterly Journal of Economics*, 131(2):799–858.

Faber, B. (2014). Trade integration, market size, and industrialization: Evidence from china's national trunk highway system. *Review of Economic Studies*, 81(3):1046–1070.

Fan, J., Tang, L., Zhu, W., and Zou, B. (2018). The alibaba effect: Spatial consumption inequality and the welfare gains from e-commerce. *Journal of International Economics*, 114:203–220.

Ju, J., Lin, J. Y., and Wang, Y. (2015). Endowment structures, industrial dynamics, and economic growth. *Journal of Monetary Economics*, 76:244–263.

Lin, J. Y. (2003). Development strategy, viability, and economic convergence. *Economic Development and Cultural Change*, 51(2):277–308.

Lin, J. Y. (2011). New structural economics: A framework for rethinking development. *World Bank Research Observer*, 26(2):193–221.

Lin, J. Y. (2012). *New Structural Economics: A Framework for Rethinking Development and Policy*. World Bank, Washington, DC.

Lin, J. Y. and Monga, C. (2010). Growth identification and facilitation: The role of the state in the dynamics of structural change. Policy Research Working Paper 5313, World Bank.

Lin, J. Y. and Wang, X. (2023). State enabling and comparative advantages. Working Paper E2023001, Institute of New Structural Economics, Peking University.

Miller, R. E. and Blair, P. D. (2009). *Input–Output Analysis: Foundations and Extensions*. Cambridge University Press, Cambridge, 2 edition.

Moretti, E. (2010). Local multipliers. *American Economic Review*, 100(2):373–377.

Romalis, J. (2004). Factor proportions and the structure of commodity trade. *American Economic Review*, 94(1):67–97.

A Proofs

A.1 Consumer access and channel retention

Lemma 1 (Consumer access)**.** *Under equations* (2) *and* (3)*,*

$$s'(z) = (\eta - 1)s(z)[1 - s(z)] > 0 \quad \textit{and} \quad \frac{d \ln P_M(z)}{dz} = -s(z) < 0.$$

If $\ell_A > \ell_P$, then $\omega'_C(z) < 0$ and $\Lambda(z) > 0$.

Proof. Write:

$$x(z) = a_P p_P(z)^{1-\eta}, \qquad a = (1 - a_P)p_A^{1-\eta}.$$

Then $s = x/(a + x)$. Because $d \ln p_P/dz = -1$,

$$\frac{x'(z)}{x(z)} = -(1 - \eta) = \eta - 1.$$

Therefore:

$$s' = \frac{x'a}{(a + x)^2} = (\eta - 1)\frac{x}{a + x}\frac{a}{a + x} = (\eta - 1)s(1 - s) > 0.$$

For the price index:

$$\ln P_M = \frac{1}{1 - \eta}\ln(a + x).$$

Hence:

$$\frac{d \ln P_M}{dz} = \frac{1}{1 - \eta}\frac{x'}{a + x} = \frac{\eta - 1}{1 - \eta}s = -s.$$

Finally:

$$\omega'_C(z) = (\ell_P - \ell_A)s'(z).$$

It is negative when $\ell_A > \ell_P$. Since $D > 0$, the definition in equation (9) then implies $\Lambda > 0$. □

At the limits $s \to 0$ and $s \to 1$, $s' \to 0$. The platform continues to reduce P_M, but the expenditure-reallocation margin vanishes at either channel boundary. This is why capture drag is largest at an intermediate platform share rather than at complete platform penetration.

A.2 Endowment-conditioned production costs

The background technology in equation (11) implies:

$$w_i = (1 - \theta_0)\frac{Y_{i0}}{L_i}, \qquad r_i = \theta_0\frac{Y_{i0}}{K_i}.$$

With $Y_{i0} = A_0 K_i^{\theta_0} L_i^{1-\theta_0}$, division by the relevant factor endowment yields equation (12). Substituting those factor prices into the candidate industry's Cobb–Douglas unit expenditure function gives:

$$
\begin{aligned}
c_{ij}^P &= \frac{1}{A_j}\left(\frac{\theta_0 A_0 e_i^{\theta_0-1}}{\theta_j}\right)^{\theta_j}\left(\frac{(1-\theta_0)A_0 e_i^{\theta_0}}{1-\theta_j}\right)^{1-\theta_j} \\
&= \frac{A_0}{A_j}\left(\frac{\theta_0}{\theta_j}\right)^{\theta_j}\left(\frac{1-\theta_0}{1-\theta_j}\right)^{1-\theta_j} e_i^{\theta_0-\theta_j} \\
&\equiv \xi_j e_i^{\theta_0-\theta_j}.
\end{aligned}
$$

Thus:

$$
\frac{\partial \ln c_{ij}^P}{\partial \ln e_i} = \theta_0 - \theta_j.
$$

For $\theta_j > \theta_0$, capital abundance lowers the candidate industry's production cost relative to the background opportunity cost. LCA still requires the external-reference double ratio in equation (15); the derivative is only the local structural component of that comparison.

A.3 Proof of Proposition 1

Proof. For a given organization r, equation (23) becomes:

$$
\pi_r(z, G) = k\mathcal{M}_0 e^{\varepsilon z + \Omega(z) G}(c + d_r)^{1-\sigma} - F_r.
$$

It is strictly increasing in G whenever $\Omega(z) > 0$. The unique raw zero-profit threshold is equation (26). Direct differentiation gives:

$$
\frac{\partial G_r^0}{\partial c} = \frac{\sigma - 1}{\Omega(z)(c + d_r)} > 0.
$$

The minimum of two strictly increasing functions is strictly increasing: for $c_2 > c_1$, both $G_P^0(c_2) > G_P^0(c_1)$ and $G_L^0(c_2) > G_L^0(c_1)$, so their minimum at c_2 exceeds their minimum at c_1. The truncation at zero preserves weak monotonicity. At an interior threshold, the relationship is strict.

Holding the external-region and background-industry ratios fixed, $\partial c/\partial \mathcal{C}_{ij}^P > 0$. The chain rule therefore gives:

$$
\frac{\partial G_E}{\partial \mathcal{C}_{ij}^P} = \frac{\partial G_E}{\partial c}\frac{\partial c}{\partial \mathcal{C}_{ij}^P} > 0
$$

at an interior threshold.

For the ACA interpretation, the focal region's delivered marginal cost is $\tau_O(z, G)(c + d_r)$. Since

$$
\frac{\partial \ln \tau_O(z, G)}{\partial G} = -(\gamma + \chi z) < 0,
$$

infrastructure lowers the focal region's total-cost ratio relative to a fixed reference region. Stronger LCA starts from a lower production-cost ratio, so it requires weakly less G both to achieve a total-cost advantage and to cover the fixed cost. The first condition is ACA; the second is viability.

Entry and sales are realization. $\square$

B Complete Entry and Organization Classification

Let:

$$b_P = k(c + d_P)^{1-\sigma}, \qquad b_L = k(c + d_L)^{1-\sigma},$$

so $b_L > b_P > 0$. It is convenient to express the state problem in terms of effective market scale:

$$X(z, G) = \mathcal{M}_0 e^{\varepsilon z + \Omega(z) G}.$$

Profits are:

$$\pi_P = X b_P - f, \qquad \pi_L = X b_L - f - F.$$

The corresponding market-scale thresholds are:

$$X_P = \frac{f}{b_P}, \qquad X_{L0} = \frac{f + F}{b_L}, \qquad X_X = \frac{F}{b_L - b_P}.$$

Lemma 2 (Three-region condition)**.** *Define:*

$$\bar{F} = f \frac{b_L - b_P}{b_P}.$$

If $F > \bar{F}$, then:

$$X_P < X_{L0} < X_X,$$

equivalently $G_P^0 < G_L^0 < G_X$, and the equilibrium states are:

$$\begin{cases} 0, & X < X_P, \\ P, & X_P \le X < X_X, \\ L, & X \ge X_X. \end{cases}$$

If $F = \bar{F}$, all three thresholds coincide. If $F < \bar{F}$, the platform-dependent interval is absent and the producer enters through L.

Proof. First:

$$\begin{aligned} X_P < X_{L0} &\iff \frac{f}{b_P} < \frac{f + F}{b_L} \\ &\iff f(b_L - b_P) < F b_P \\ &\iff F > \bar{F}. \end{aligned}$$

Second:

$$X_{L0} < X_X \iff \frac{f+F}{b_L} < \frac{F}{b_L - b_P}$$
$$\iff f(b_L - b_P) < F b_P$$
$$\iff F > \bar{F}.$$

Thus the same condition orders all three thresholds. Below X_P, both profits are negative. At X_P, $\pi_P = 0$, while:

$$\pi_L - \pi_P = X_P(b_L - b_P) - F = \bar{F} - F < 0.$$

Because:

$$\pi_L - \pi_P = X(b_L - b_P) - F$$

is strictly increasing in X, it crosses zero exactly once at X_X. This proves the three states. Equality collapses all thresholds to one point. If $F < \bar{F}$, the local mode is already more profitable at the first viable entry scale, so P is never selected. □

C Proof of Proposition 2

Proof. Define:

$$\Delta(c) = (c + d_L)^{1-\sigma} - (c + d_P)^{1-\sigma} > 0.$$

The raw switch threshold is:

$$G_X = \frac{\ln F - \ln[k\mathcal{M}_0 e^{\varepsilon z}] - \ln \Delta(c)}{\Omega(z)}.$$

Its cost derivative is:

$$\frac{\partial G_X}{\partial c} = -\frac{\Delta'(c)}{\Omega(z)\Delta(c)}.$$

Because:

$$\Delta'(c) = (1-\sigma)\left[(c + d_L)^{-\sigma} - (c + d_P)^{-\sigma}\right] < 0,$$

we have $\partial G_X/\partial c > 0$. Truncation at zero preserves weak monotonicity. The LCA result follows from $\partial c/\partial \mathcal{C}^P_{ij} > 0$.

For $G > G_X$:

$$\pi_L - \pi_P = k\mathcal{M}_0 e^{\varepsilon z + \Omega G}\Delta(c) - F > 0.$$

Hence L is selected whenever it also weakly dominates no entry; the classification in Lemma 2 covers the possible entry orders.

Since $d_L < d_P$:

$$(c + d_L)^{1-\sigma} > (c + d_P)^{1-\sigma},$$

and therefore $\mathcal{R}_L > \mathcal{R}_P$ at a common (z, G). The payment-flow definitions imply:

$$\rho_L = 1 > 1 - \frac{d_P}{\mu(c + d_P)} = \rho_P.$$

Both inequalities are strict, so:

$$B_j^L = \rho_L \mathcal{R}_L > \rho_P \mathcal{R}_P = B_j^P.$$

The external payment changes from $d_P q_P > 0$ to zero in the polar baseline, while the unit organizational service $d_L q_L$ and the shared fixed organizational service F are locally supplied. The organizational switch therefore changes local service income, external payments, and total first-round local income. □

D Proof of Proposition 3

D.1 Income derivatives within a producer state

Fix an active producer organization r. The local producer-income term can be written:

$$B_j^r(z, G) = A_r \exp\{\varepsilon z + \Omega(z) G\},$$

where:

$$A_r = \rho_r \mathcal{M}_0 \mu^{1-\sigma} (c + d_r)^{1-\sigma} > 0.$$

Therefore:

$$\frac{\partial \ln B_j^r}{\partial G} = \Omega(z) > 0,$$

and:

$$\frac{\partial \theta_r}{\partial G} = \Omega(z) \theta_r (1 - \theta_r) > 0.$$

Define:

$$a(G) = \varepsilon + (\sigma - 1)\chi G \geq 0.$$

The nominal integration effect within organization r is:

$$h_Y(G) = \theta_r(G) a(G) - \Lambda(z).$$

Its derivative is:

$$h_Y'(G) = \Omega(z)\theta_r(1 - \theta_r) a(G) + \theta_r(\sigma - 1)\chi > 0. \tag{57}$$

The price term $\alpha_M s(z)$ is independent of G, so:

$$h_R'(G) = h_Y'(G) > 0$$

within an active state. In the no-entry state, $\theta = 0$; both effects are constant in G until entry.

D.2 Jumps at state changes

Immediately before entry, $\theta = 0$, so $h_Y = -\Lambda$. Immediately after entry, the viable mode generates positive sales and local producer income, so $\theta > 0$. Since $a(G) \geq 0$, the integration effect jumps weakly upward and strictly upward if $a(G) > 0$.

At a P-to-L switch, Proposition 2 gives $B_j^L > B_j^P$. Hence $\theta_L > \theta_P$, while $a(G)$ and $\Lambda(z)$ are unchanged at the switch. Both nominal and real effects jump strictly upward.

D.3 Existence, uniqueness, and ordering

Completion of the proof. The preceding derivatives and jump results imply that the integration effects are globally nondecreasing in G, strictly increasing in every active organization, and have no downward state jumps. Under the baseline $\ell_A > \ell_P$ and an interior platform share, the no-entry effect is strictly negative because $\Lambda > 0$.

As $G \to \infty$, the local mode is selected, $\theta_L \to 1$, and:

$$a(G) = \varepsilon + (\sigma - 1)\chi G \to \infty$$

when $\chi > 0$. Hence $h_Y(G) \to \infty$, and the real effect is larger still. More generally, it is sufficient to assume a finite upper endpoint at which the relevant effect is positive. Monotonicity and endpoint crossing therefore imply a unique minimum threshold, allowing the crossing to occur either continuously or at an upward jump.

Pointwise:

$$h_R(G) = h_Y(G) + \alpha_M s(z) > h_Y(G).$$

Thus every G that makes the nominal effect nonnegative also makes the real effect positive, establishing $G_R \leq G_Y$.

Suppose both roots lie in the same active continuous organization and are interior. At G_Y, $h_Y(G_Y) = 0$, so:

$$h_R(G_Y) = \alpha_M s(z) > 0.$$

Since h_R is continuous and strictly increasing in that organization, its zero must occur strictly below G_Y. Hence $G_R < G_Y$. If both effects jump from negative to nonnegative at the same discrete state change, their minimum thresholds are identical even though $h_R > h_Y$ on either side.

It remains to establish the comparative static with respect to LCA. For the local mode:

$$B_j^L = \mathcal{M}\mu^{1-\sigma}(c + d_L)^{1-\sigma},$$

which is strictly decreasing in c. For the platform mode:

$$B_j^P = \mathcal{M}\mu^{1-\sigma}\left[(c+d_P)^{1-\sigma} - \frac{d_P}{\mu}(c+d_P)^{-\sigma}\right].$$

Differentiating the bracket gives:

$$\begin{aligned}(1-\sigma)(c+d_P)^{-\sigma} &+ \frac{\sigma d_P}{\mu}(c+d_P)^{-\sigma-1} \\ &= -(\sigma-1)c(c+d_P)^{-\sigma-1} < 0,\end{aligned}$$

where $\mu = \sigma/(\sigma-1)$. Thus lower c raises local producer income in either mode. It also weakly advances entry and the switch to the higher-income local mode by Propositions 1 and 2. At every fixed G, stronger LCA therefore weakly raises θ_{r^*} and the two integration effects. Their minimum thresholds weakly fall. The relationship is strict when the relevant root remains interior to the same active organizational state. $\square$

E Proof of Proposition 4

Proof. Define the infrastructure provider's first-order condition:

$$\Phi(G,\vartheta) = \mathcal{U}_I'(G) - \kappa_I(\vartheta)C'(G) = 0.$$

Its derivative with respect to G is:

$$\Phi_G = \mathcal{U}_I''(G) - \kappa_I(\vartheta)C''(G) < 0.$$

The objective is therefore strictly concave and has at most one interior solution. Under the maintained endpoint conditions, that solution exists and is unique. Since:

$$\Phi_\vartheta = -\kappa_I'(\vartheta)C'(G) > 0,$$

implicit differentiation gives:

$$G'(\vartheta) = -\frac{\Phi_\vartheta}{\Phi_G} > 0.$$

For any fixed structural threshold G_H, the monotonicity of $G(\vartheta)$ implies that the set $\{\vartheta : G(\vartheta) \geq G_H\}$ is an interval. When nonempty, it has a unique minimum ϑ_H, except for an economically irrelevant flat boundary if the infrastructure solution itself is constrained. Increasing ϑ reduces $G_H - G(\vartheta)$. Stronger LCA lowers G_E, G_L, G_R, G_Y by Propositions 1–3; applying the increasing inverse of $G(\vartheta)$ weakly lowers the associated participation thresholds.

For the protection comparison, a consumer wedge $\tau_C > 1$ changes the effective platform price to $\tau_C \bar{p}_P e^{-z}$. The CES price index is increasing in an input price, so protection raises P_M and lowers the platform expenditure share. On the producer side:

$$\frac{\partial G_P^0}{\partial d_P} = \frac{\sigma - 1}{\Omega(z)(c + d_P)} > 0.$$

Thus protection makes the low-fixed-cost external entry mode harder for a marginal producer.

Let:

$$\Delta = (c + d_L)^{1-\sigma} - (c + d_P)^{1-\sigma}.$$

Because:

$$\frac{\partial \Delta}{\partial d_P} = (\sigma - 1)(c + d_P)^{-\sigma} > 0,$$

the switch threshold satisfies:

$$\frac{\partial G_X}{\partial d_P} = -\frac{1}{\Omega(z)\Delta}\frac{\partial \Delta}{\partial d_P} < 0.$$

Protection can therefore induce a switch toward L, but it does so by worsening P, while simultaneously harming consumer access and making platform-based entry more difficult. Facilitation raises G, which increases market access under both modes and does not enter the consumer price equation. The policies are not equivalent. □

F Boundary Cases and Counterexamples

F.1 No consumer-side capture gap

If $\ell_A = \ell_P$, then:

$$\omega_C(z) = \ell_A, \qquad \Lambda(z) = 0.$$

The nominal effect becomes:

$$\frac{d \ln Y}{dz} = \theta_r[\varepsilon + (\sigma - 1)\chi G] \geq 0$$

in every active producer state. A negative nominal effect can then arise only in an extension with another first-round loss. This case confirms that the baseline negative region is generated by an explicit payment-incidence gap rather than platform terminology.

If $\ell_P > \ell_A$, then $\Lambda < 0$. Platform substitution raises consumer-side retention as well as reducing the price index. Both nominal and real effects shift upward; the low-infrastructure loss region may vanish.

F.2 No producer-side spatial payment gap

Setting $\lambda = 1$ in equation (35) gives $\rho_P = 1 = \rho_L$. The organization switch can still raise sales because $d_L < d_P$, but it no longer changes the spatial retention rate. This case separates the cost advantage of local organization from its payment-location advantage.

F.3 No infrastructure–platform complementarity

If $\chi = 0$, producer market access is:

$$\mathcal{M}(z, G) = \mathcal{M}_0 e^{\varepsilon z + (\sigma - 1)\gamma G}.$$

Infrastructure can still raise the level of candidate-industry income and therefore θ_r, but it no longer increases the marginal producer benefit of platform integration directly. Equation (42) loses the term $(\sigma - 1)\chi G$. The threshold mechanism can survive, but the strong complementarity interpretation cannot.

F.4 Collapse of the platform-dependent region

Lemma 2 gives the exact counterexample to a universal three-stage development path. When $F \le \bar{F}$, the local mode is sufficiently inexpensive in fixed-cost terms that the candidate industry enters directly through L. Platform dependence is therefore neither necessary nor universal.

F.5 Weak rather than strict income-threshold ordering

Suppose an entry or organizational switch occurs at $\widehat{G}$, with:

$$h_R(\widehat{G}^-) < 0, \qquad h_Y(\widehat{G}^-) < 0,$$

and:

$$h_R(\widehat{G}^+) \ge 0, \qquad h_Y(\widehat{G}^+) \ge 0.$$

Then:

$$G_R = G_Y = \widehat{G}.$$

The pointwise inequality $h_R > h_Y$ is not enough to create two different minimum thresholds when a common discrete jump crosses both zeroes. This is why Proposition 3 states strict ordering only under a continuous interior-root condition.

F.6 Instability of the local-income closure

If:

$$D(z) = 1 - \beta - \alpha_M \omega_C(z) \le 0,$$

the household-spending recursion does not converge. This is not an alternative equilibrium of the paper. It is an inadmissible parameter region. The baseline values satisfy $D(z) > 0$ for the full numerical support.

G Mapping to the Companion EL Income Block

The companion access–capture model can be written:

$$Y^{EL} = \frac{B^{EL} + \omega^{EL} G^{EL}}{1 - \beta^{EL} - \alpha^{EL} \omega^{EL}}.$$

The present paper does not insert this expression into the producer problem. It uses the following one-to-one mapping only after first-round payments have been determined:

Table 4: Mapping to the companion income-propagation block

Companion object	Present object	Accounting rule
$B^{EL} + \omega^{EL} G^{EL}$	$B_0 + B_j^{r^*}$	autonomous first-round local income only
α^{EL}	α_M	household market-good expenditure share
β^{EL}	β	locally supplied nontraded-good expenditure share
ω^{EL}	$\omega_C(z)$	local income content of induced market-good spending
consumer price index	$P_M(z)$	identical CES market-good price concept
nominal income	Y	equation (39)
consumption-equivalent income	R	equation (40)

The candidate-industry sales payment enters $B_j^{r^*}$ once. It does not also enter ω_C, which applies only to subsequent household market-good expenditure. The external platform organization payment enters ρ_P once. Consumer-channel external payments enter ℓ_P and ω_C, not ρ_P. This separation prevents production, organizational services, and induced household spending from being counted twice.

H Numerical Reproducibility and Mechanism Closures

The numerical implementation consists of:

- `code/numerical/model.py`, which implements the equations and equilibrium choices;
- `code/numerical/generate_figures.py`, which generates all figures and source-data CSV files;
- the validation script, which checks analytical derivatives, finite differences, threshold uniqueness, the three-region condition, boundary cases, and state monotonicity.

For $z = 0.5, c = 0.8$, the code obtains:

$$G_P^0 = -0.070176, \quad G_L^0 = 0.126129, \quad G_X = 0.256065,$$

$$G_R = 0.046348, \quad G_Y = 0.550487.$$

Finite-difference derivatives have the signs established in the proofs. The program also verifies that lowering F below $\bar{F}$ removes the platform-dependent interval and that equal consumer-channel local contents set Λ exactly to zero.

The file `figures/source_data/appendix_mechanism_closures.csv` reports the full G-grid for:

1. the baseline;
2. $\ell_A = \ell_P$;
3. complete platform-service localization;
4. $\chi = 0$;
5. partial platform-service localization.

All reported exercises use normalized, illustrative values. No parameter is estimated, and the exercises do not constitute a calibration.